

Data Flow Diagram Specification

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	2 Marks

Data Flow Diagram (Level-1 DFD):

Illustrates the logical movement of data between client entities, application processes, database stores, and external AI services.

Symbol	DFD Element	System Component in AI-FitTrack-API	Function
Square	External Entity	User / API Client (Thunder Client, Web/Mobile)	Issues HTTP requests and consumes JSON responses
Rounded Box	Process	Processes 1.0 - 5.0 (Auth, CRUD, Search, AI Rec, Insights)	Executes Express.js routing, controller logic, and validation
Open Bar	Data Store	D1: Users Store, D2: Workouts Store (MongoDB)	Persists User and Workout documents with indexing
Double Square	External Service	Google Gemini 1.5 Flash Generative AI Service	Receives contextual prompts and returns structured routines

AI-FitTrack-API : Level-1 Data Flow Diagram (DFD)

Information Flow Between Client, REST Processes, MongoDB Data Stores & Google Gemini AI

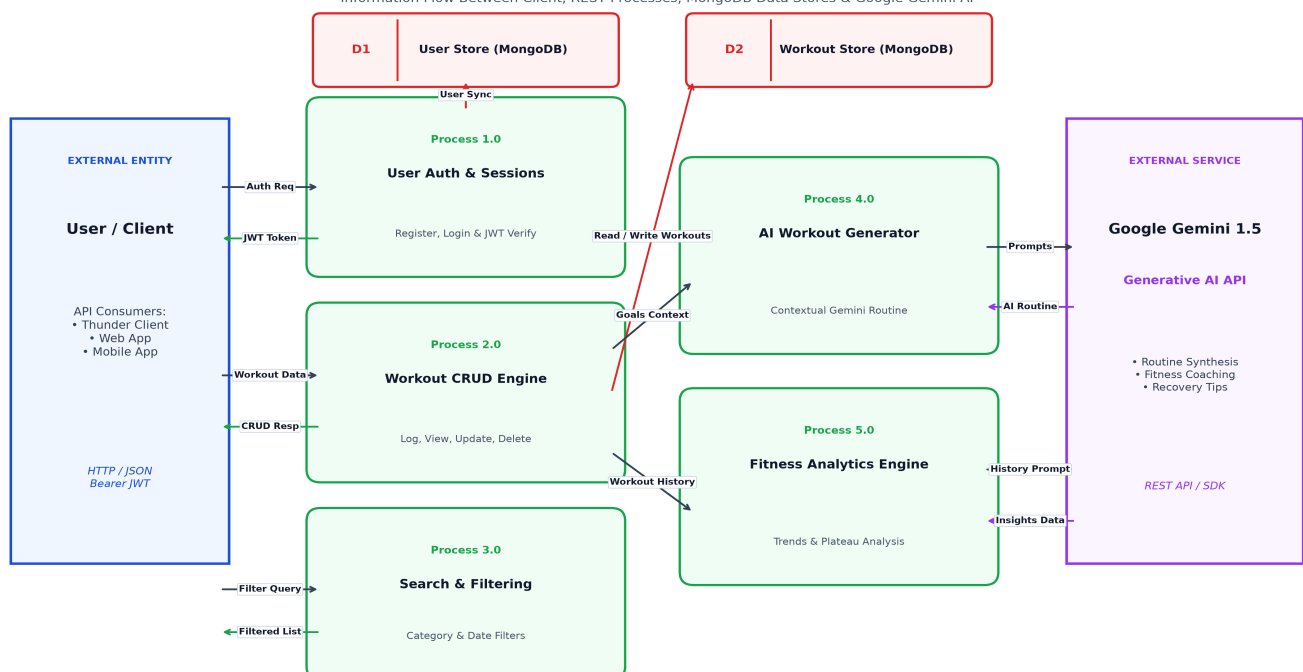


Figure 1: Level-1 Data Flow Diagram of AI-FitTrack-API Architecture